



CUBAN NATIONAL CHEMISTRY CONTEST 2022

"You should never be afraid of what you're doing, when it's right."

Marie Curie

Name(s): _____ Last name(s): _____

School: _____

Municipality: _____ Province: _____

CODE: _____ Test: **DAY 2. TENTH GRADE EXAM**

Instructions

- Write clearly all your data in the appropriate place (names, last names, school, municipality, province). Do not write anything in the field CODE.
- Verify that your questions booklet is complete. This is the Day 2 problem set, with an individual exam for 10th grade. The exam has 2 problems (problem 5, 6 items; problem 6, 3 items) and a total of 5 pages.
- All data, constants, conversions and basic equations needed are included in the exam. You are allowed to use a scientific calculator.
- At the beginning of each problem its scoring (from a total of 100 points), and each item scoring (in marks) is indicated.

Success!



Ministerio de Educación
República de Cuba



CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Equation of ideal gas	$p \cdot V = n \cdot R \cdot T$
Gases constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta E + W$
Zero on Celsius scale	273,15 K	Gibbs' free energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp\left(-\frac{E_A}{R \cdot T}\right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$	$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

PERIODIC TABLE OF ELEMENTS

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1 H 1.008	2											13	14	15	16	17	2 He 4.003
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3	4	5	6	7	8	9	10	11	12	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.97	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.95	43 Tc -	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57-71	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po -	85 At -	86 Rn -
87 Fr -	88 Ra -	89-103	104 Rf -	105 Db -	106 Sg -	107 Bh -	108 Hs -	109 Mt -	110 Ds -	111 Rg -	112 Cn -	113 Nh -	114 Fl -	115 Mc -	116 Lv -	117 Ts -	118 Og -

Problem 5. Sulfur chemistry (30 points)

5.1	5.2	5.3	5.4	5.5	5.6	Total	Total
9	23	4	4	15	5	60	30

As for other non-metals, the existence of a wide variety of compounds with sulfur-halogen bonds has been demonstrated, where some play a fundamental role in the synthesis of organic or inorganic substances. The chemical synthesis of some sulfur halides or oxohalides is described below:

- i) Sulfur hexafluoride, SF_6 , at STP is a gas obtained by combustion of elemental sulfur in a fluorine atmosphere.
- ii) The S_2F_{10} can be obtained in a liquid state under similar conditions to the previous ones, but only if temperature and pressure are carefully controlled.
- iii) Sulfur tetrafluoride vapors, SF_4 , are obtained together with SF_6 , as products of the S_2F_{10} disproportionation when heated to 150°C .
- iv) At STP the gaseous compounds SF_5Cl and SF_5Br are formed by the treatment of the S_2F_{10} with chlorine and bromine respectively.
- v) SOF_4 vapors are obtained by air oxidation of sulfur tetrafluoride.
- vi) Sulphoryl fluoride, SO_2F_2 , at STP is a gas formed by the reaction of sulfur dioxide with fluorine.
- vii) The S_2Cl_2 is a liquid substance at STP with a structure similar to hydrogen peroxide. This compound is obtained from the combustion of sulfur in chlorine atmosphere.
- viii) Liquid sulfur dichloride, SCl_2 , is formed by post-treatment of S_2Cl_2 with chlorine.
- ix) Liquid thionyl chloride, SOCl_2 , is industrially prepared by the reaction of sulfur trioxide with sulfur dichloride in chlorine atmosphere.

5.1 Write the equations for the nine reactions described above.

5.2 What is the most probable geometry adopted by the domains (bonding or non-bonding pairs of electrons) around the sulfur atoms in each of the obtained compounds? Draw the corresponding structures according to the VSEPR theory. What molecules have zero net dipole moment?

NOTE: For each compound, write in the corresponding box the chemical structure, the type of geometry and whether or not the molecule has a zero net dipole moment (write Yes or No as appropriate).

Chemical formula	i) SF ₆	ii) S ₂ F ₁₀	iii) SF ₄	iv) SF ₄ X (X = Cl, Br)
Chemical structure				
Geometry				
$\mu = 0$				
v) SOF ₄	vi) SO ₂ F ₂	vii) S ₂ Cl ₂	viii) SCl ₂	ix) SOCl ₂

5.3 The higher reactivity of the S₂F₁₀ respect to sulfur hexafluoride makes it a suitable starting material for synthesis of other halogenated sulfur compounds. What features of their structures lead to this difference in reactivity?

5.4 Compare the S-X bond length and X-S-F bond angle of the SF₅X compounds, X = F, Cl, Br.

NOTA: NOTE: Write the chemical formula of the SF₅X compounds in such a way that there is an increasing order of magnitude.

5.5 Sulfur tetrafluoride forms adducts with boron trifluoride which can be represented in ionic form as [SF₃][BF₄]. Write the corresponding chemical equation. In this reaction, indicate the compounds that exhibit Lewis acid and base properties, respectively. For the reactants and the ionic species of the adduct formed, write the chemical structure and electronic distribution of the central atom according to the hybridization theory developed by Pauling.

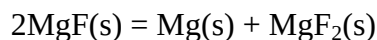
5.6 In addition to applications in the synthesis of organic compounds, thionyl chloride is used industrially to prepare anhydrous metal halides like BiCl₃. What is the simplified electronic distribution of bismuth? Explain why bismuth chlorides with higher oxidation states are not known.

Problem 6. Magnesium thermochemistry (20 points)

6.1	6.2	6.3	Total	Total
13	20	7	40	20

It is generally possible to consider that a given compound is thermodynamically unstable respect to decomposition into its constituent elements or respect to disproportionation, if either of these chemical transformations involves a large release of energy.

6.1 Show by calculations that the unusual magnesium(I) fluoride is a thermodynamically unstable compound respect to its disproportionation at 298 K:



Note that the lattice energies of MgF_2 and MgF are 2952 and 930 kJ/mol, respectively. The sublimation heat of Mg(s) is 147.7 kJ/mol and the difference between the primary and secondary ionization energy of this metal is 713 kJ/mol. The above data were determined or estimated at STP.

The formation energy of magnesium(II) fluoride was determined in 1964 by the direct reaction of the elements in a calorimeter. Combustion in fluorine atmosphere of a 0.50000 g sample of magnesium in the calorimeter causes the temperature to rise by 1.44093°C. The heat capacity of the calorimeter is 3433.46 cal/°C. However, as the reaction between both elements is very vigorous, a 195.8 g nickel crucible and a 304.47 g compacted magnesium(II) fluoride coating were also placed inside the calorimeter. After the calorimetric measurement, the sample under study was treated with a dilute solution of hydrochloric acid, producing 36.9 cm³ of hydrogen under normal conditions.

6.2 What is the value of the formation energy of 1 mole of magnesium(II) fluoride?

6.3 Is magnesium(I) fluoride a thermodynamically stable salt respect to decomposition into its constituent elements? Use the information you obtained in the previous exercises.

If you could not solve parts 6.1 and 6.2, consider $\Delta H_f^\circ(\text{MgF}_2) = -1000$ kJ/mol and $\Delta H^\circ(\text{Reaction 1}) = -500$ kJ/mol

Data:

$V_m = 22.4$ L/mol in normal conditions

$c_p(\text{Ni}) = 6.21$ cal/(mol·°C) and $c_p(\text{MgF}_2) = 14.72$ cal/(mol·°C)

1 cal = 4.184 J